

CSCN8020 Assignment 1 — Reinforcement Learning Programming

Student Name Anthony Izevbokun
Student ID 9016626

100-Word Summary of Work Completed

This assignment addresses four reinforcement learning tasks within a single Jupyter Notebook. The first problem models a pick-and-place robot as a Markov Decision Process, specifying states, actions, transitions, and a reward function designed to encourage fast and smooth operation. The second problem manually applies two Bellman optimality updates to a 2x2 Gridworld and confirms the results through code. For the 5x5 Gridworld, both synchronous and in-place value iteration are implemented and compared by convergence speed and runtime. Finally, off-policy Monte Carlo control with weighted importance sampling is applied to the same grid, and its value estimates are benchmarked against the dynamic programming solution using episode count, runtime, and mean error metrics.

Public GitHub Repository

https://github.com/AntUse/CSCN8020_Assignment1.git